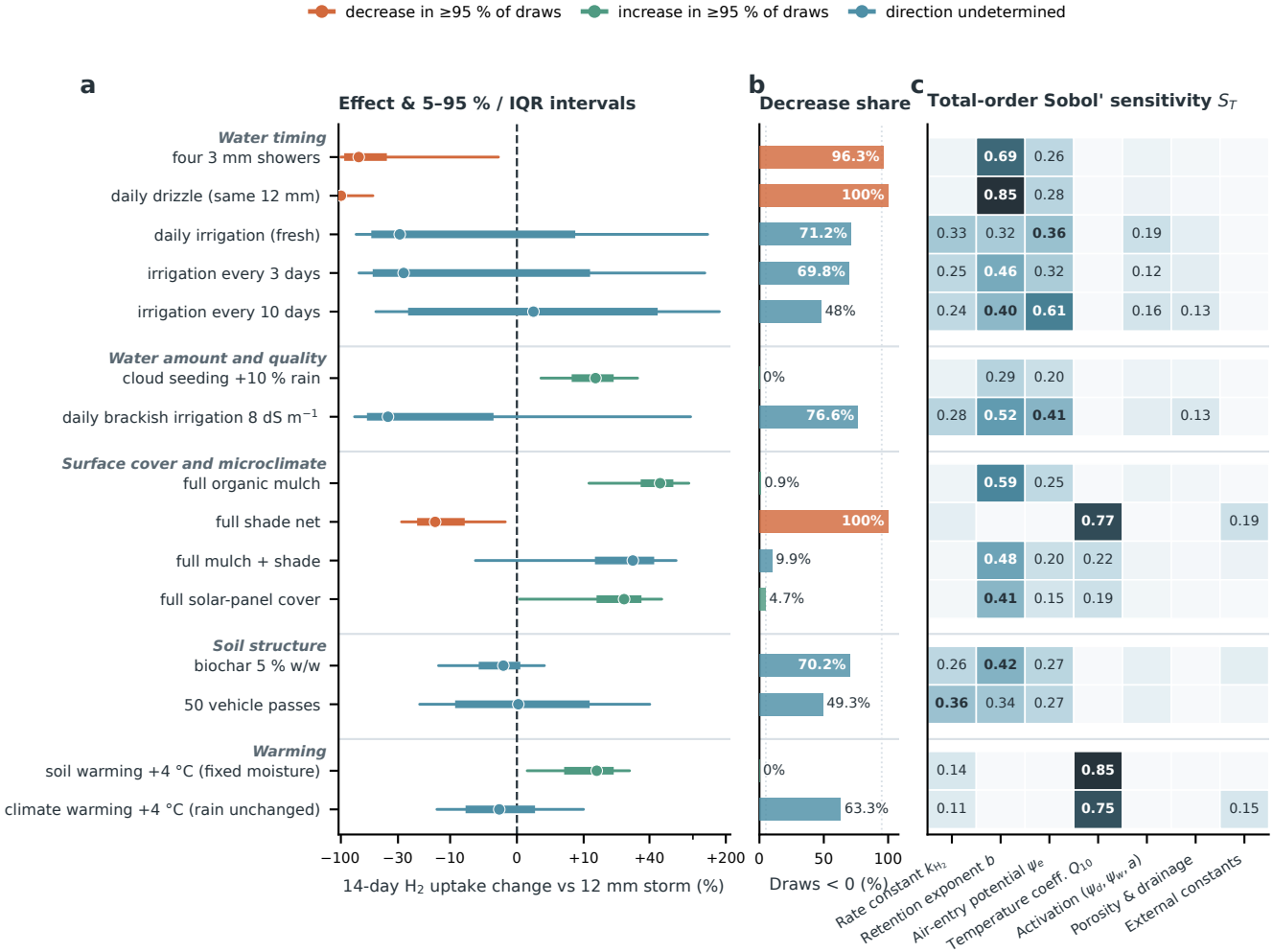

decrease in ≥95 % of draws
increase in ≥95 % of draws
direction undetermined
a
Effect & 5–95 % / IQR intervals
Water timing
four 3 mm showers
daily drizzle (same 12 mm)
daily irrigation (fresh)
irrigation every 3 days
irrigation every 10 days
Water amount and quality
cloud seeding +10 % rain
daily brackish irrigation 8 dS m−1
Surface cover and microclimate
full organic mulch
full shade net
full mulch + shade
full solar-panel cover
Soil structure
biochar 5 % w/w
50 vehicle passes
Warming
soil warming +4 °C (fixed moisture)
climate warming +4 °C (rain unchanged)
−100 −30 −10 0 +10 +40 +200
14-day H2 uptake change vs 12 mm storm (%)
b
Decrease share
96.3%
100%
71.2%
69.8%
48%
0%
76.6%
0.9%
100%
9.9%
4.7%
70.2%
49.3%
0%
63.3%
0 50 100
Draws < 0 (%)
c
Total-order Sobol' sensitivity ST
0.69 0.26
0.85 0.28
0.33 0.32 0.36 0.19
0.25 0.46 0.32 0.12
0.24 0.40 0.61 0.16 0.13
0.29 0.20
0.28 0.52 0.41 0.13
0.59 0.25
0.77 0.19
0.48 0.20 0.22
0.41 0.15 0.19
0.26 0.42 0.27
0.36 0.34 0.27
0.14 0.85
0.11 0.75 0.15
Rate constant kH2
Retention exponent b
Air-entry potential ψe
Temperature coeff. Q10
Activation (ψd, ψw, a)
Porosity & drainage
External constants